

Problem B. Binary search

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

You are given a sorted array of numbers, and followed by number of queries, for each query if the queried number is present in the array print its position, else print -1.

Input

First line contains N Q, number of elements in the array and number of queries to follow.

Second line contains N numbers, the elements of the array. Each number will be $-10^9 \leq a_i \leq 10^9$, $0 < N \leq 10^5$, $0 < Q \leq 5 \cdot 10^5$

Output

For each element in the query, print the elements 0 based location of its first occurrence, if present, otherwise print -1.

Example

Input	Output
5 4 2 4 7 7 9 7 10 4 2	2 -1 1 0